Regular Expressions

Matching Text in RHEL / Linux

Simple explanation + practical grep examples

```
grep
^ERROR
error$
[0-9]
.*
```

Regex = a pattern used to find text

What is a Regular Expression?

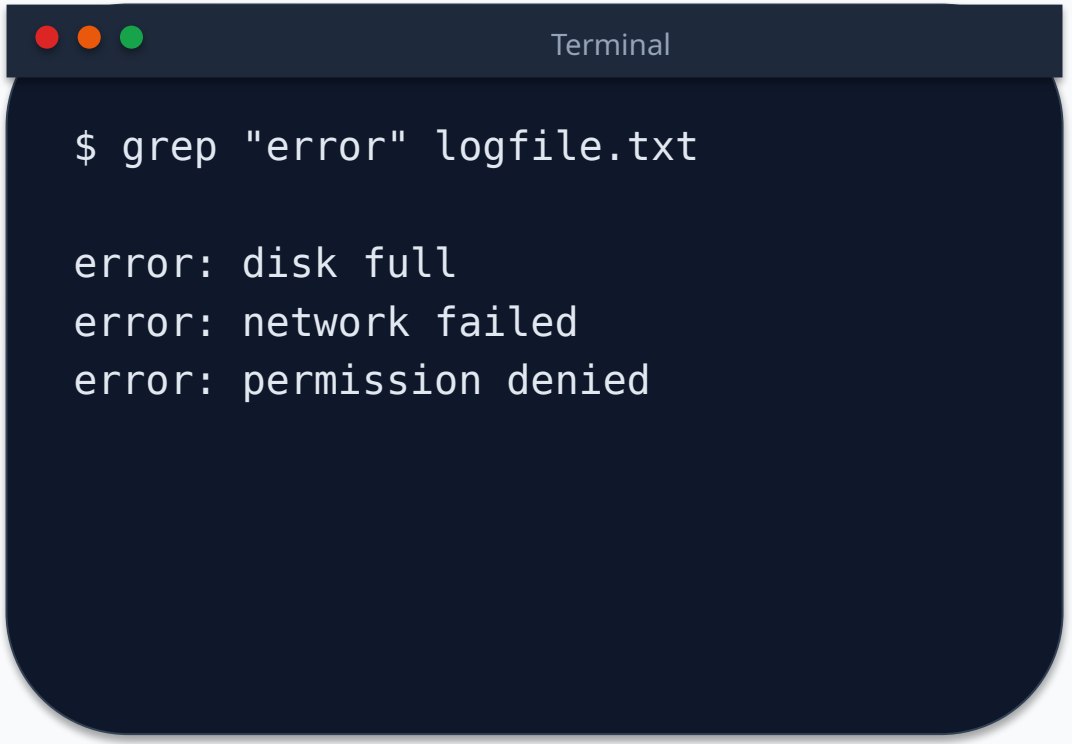
Think of regex as a smart search pattern.

A regular expression (regex) is a pattern used to find specific text.

It is more powerful than a simple word search.

In RHEL/Linux, grep is one of the most useful commands for regex.

Regex can also be used with tools such as less and vim.



Search pattern → Matching lines

Why Do We Need Regex?

Useful when files or logs contain thousands of lines.

Terminal

```
server started
user admin logged in
ERROR: disk full
user john logged in
ERROR: network failed
backup completed
ERROR: permission denied
```

We want only ERROR lines

Terminal

```
$ grep "ERROR" logfile.txt

ERROR: disk full
ERROR: network failed
ERROR: permission denied
```

The Most Important Regex Symbols

Symbol	Meaning	Example	Simple meaning
^	Beginning of line	**^ERROR**	Lines starting with ERROR
$	End of line	**ERROR$**	Lines ending with ERROR
.	Any one character	**c.t**	cat, cut, cot
*****	Zero or more	**ab***	a, ab, abb...
[ ]	One character from a grou	**[abc]at**	aat, bat, cat
[0-9]	Any digit	**[0-9]**	0 through 9

^ Means: Beginning of the Line

Use ^ when the text must appear at the start.

Terminal

```
ERROR: disk full
WARNING: memory high
system ERROR
ERROR: permission denied
```

Terminal

```
$ grep "^ERROR" logfile.txt

ERROR: disk full
ERROR: permission denied
```

"^ERROR" = line starts with ERROR

$ Means: End of the Line

Use $ when the text must appear at the end.

Terminal

```
system error
network error
error occurred
disk error
ERROR
```

Terminal

```
$ grep "error$" logfile.txt

system error
network error
disk error
```

"error$" = line ends with error

. * [ ] — Easy Examples

These symbols make your search more flexible.

`.` → any one character: c.t can match cat, cut, cot

`*` → zero or more occurrences: ab* can match a, ab, abb…

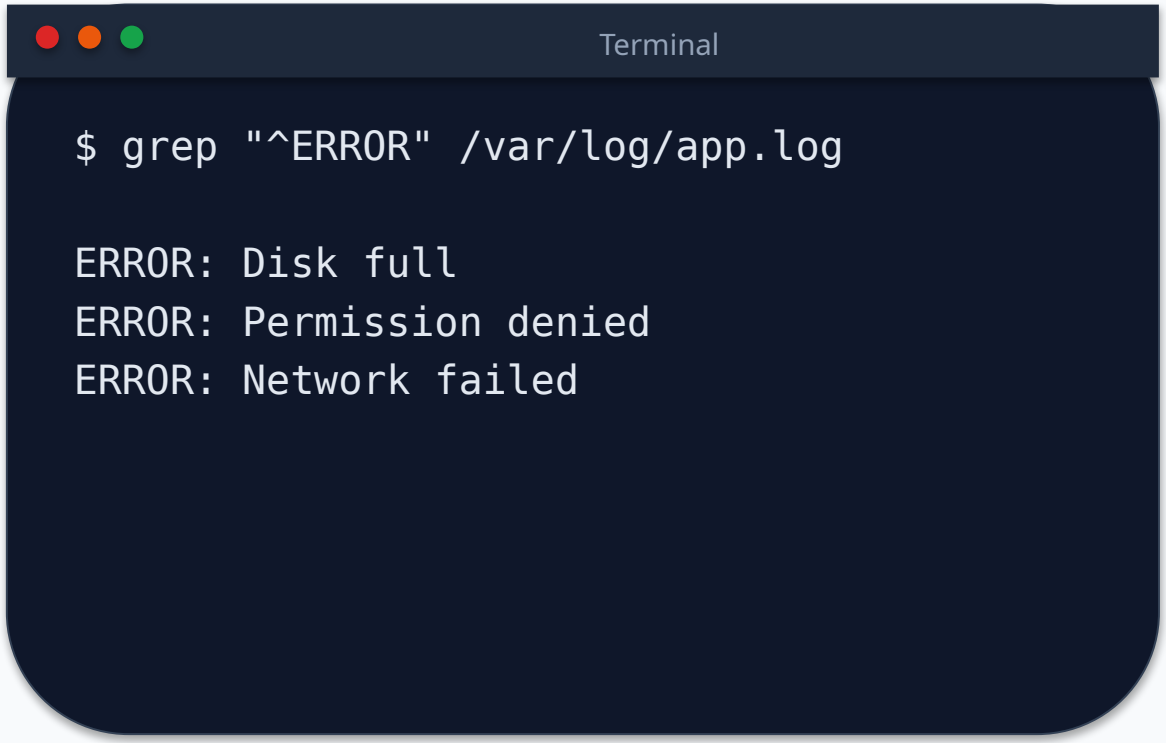
`[abc]` → one character from the group: [abc]at

`[0-9]` → any digit from 0 to 9



Real Linux Administrator Example

Finding errors in a server log.



Large log files are common in Linux administration.

Instead of reading every line, use a pattern.

`^ERROR` finds only lines that start with ERROR.

This quickly isolates problems that need attention.

Regex + grep = fast text filtering

Quick Memory Sheet

Keep these six patterns in mind.

`^` **Beginning**
`^ERROR`

`$` **Ending**
`ERROR$`

`.` **Any one character**
`c.t`

`*` **Zero or more**
`ab*`

`[ ]` **Choose one**
`[abc]at`

`[0-9]` **Any digit**
`[0-9]`

Regex = pattern → grep → matching text